

Junior Automation & Controls Technician

Portfolio Project Roadmap - College Edition Platform & Evidence Guide

PROGRAMME WINDOW	7 September 2026 - 28 February 2027
TARGET	Junior Automation & Controls Technician
LEARNING LOAD	25 weeks 17 hours/week approximately 425 hours
FOCUS	Industrial automation, instrumentation, process control and field troubleshooting

Purpose

COORDINATED LEARNING SYSTEM

The four projects are integrative performance assessments. Each one must prove engineering reasoning through requirements, design, implementation, testing, troubleshooting, documentation and an honest technical demonstration.

Prepared for Emmanuel Januarie

College-style personal learning programme - not an accredited qualification

Recommended learning stack

Skill area	Preferred platform	Purpose / fallback
PLC	TIA Portal + S7-PLCSIM	Primary S7-1200/S7-1500-style environment; use paper/design evidence if unavailable
HMI/SCADA	Ignition	Tags, views, alarms, trends/history and OPC UA
Process	Factory I/O if accessible	Use another safe simulator or a documented software model
Protocols	OPC UA + Modbus TCP/RTU	Addressing, mapping, data quality and recovery
Networking	Windows tools + Wireshark	IPv4, TCP/IP and communications diagnostics
Drawings	QElectroTech + diagrams.net	Schematics, P&IDs, loops and network diagrams
Rockwell	Studio 5000 concepts	Focused vendor translation; do not claim professional competence

PLATFORM PRINCIPLE

Siemens remains the primary hands-on PLC story, Ignition the primary SCADA story, and Rockwell a clearly labelled introductory translation exercise. Software access and licensing vary; never fabricate screenshots or implementation claims.

Official reference starting points

Siemens STEP 7 / TIA Portal: <https://www.siemens.com/en-us/products/tia-portal/step7/>

Ignition documentation: <https://docs.inductiveautomation.com/docs/8.1/>

Rockwell Studio 5000: <https://www.rockwellautomation.com/en-us/docs/studio-5000-logix-designer/>

Portfolio governance and release rules

EVERY PROJECT BEGINS WITH

- ☐ Problem statement and scope
- ☐ Assumptions and exclusions
- ☐ Safety and authorization boundary
- ☐ Architecture and interfaces
- ☐ Measurable acceptance criteria
- ☐ Versioned baseline

EVERY PROJECT ENDS WITH

- ☐ Normal/fault/recovery tests
- ☐ Defect closure and regression
- ☐ Source files or honest design substitute
- ☐ Evidence index and README
- ☐ Short technical demonstration
- ☐ Limitations and next steps

Common scoring rubric

Category	Weight	Release question
Requirements and design	15%	Is the intended behavior explicit and traceable?
Technical implementation	25%	Does the system behave correctly and remain understandable?
Fault handling and safety	20%	Are abnormal states protected and recoverable?
Testing and evidence	20%	Do expected and actual results prove the claims?
Documentation	10%	Can another technician review and maintain it?
Demonstration	10%	Can the learner explain decisions and limitations?

RELEASE THRESHOLD

Target at least 75/100 with no category below 60%. Any uncorrected safety, ownership, stale-data or protective-response defect blocks release.

Project 1 - Industrial Motor & Conveyor Control Panel

BUILD WINDOW	PROJECT LABEL
5 Oct - 15 Nov 2026	Simulation / Training Project

PROJECT BRIEF

Design and simulate a conveyor control panel that demonstrates electrical tracing, S7-style PLC logic, motor/VFD interfaces, modes, interlocks, sequencing and fault recovery.

Learning objectives

- ☐ Translate a written process into I/O, logic and tests
- ☐ Demonstrate Ladder, timer, counter and latching patterns
- ☐ Make command ownership and protective conditions visible
- ☐ Diagnose at least five faults by layer

Platforms and tools

PREFERRED	ACCESS RULE
☐ Siemens TIA Portal + S7-PLCSIM	☐ Confirm software availability before committing
☐ Factory I/O if available	☐ Use official documentation
☐ QElectroTech or equivalent	☐ Record version used
☐ Spreadsheet/document tools	☐ Use a documented fallback if unavailable
	☐ Do not overstate vendor competence

Functional requirements

- ☐ Start/Stop and controlled restart
- ☐ Auto/Manual and Local/Remote simulation
- ☐ Two conveyor zones and object sensors
- ☐ Motor command, running feedback and overload
- ☐ VFD ready, run, speed and fault interface
- ☐ Jam timer, count target, alarms and reset
- ☐ Overview, faceplate and diagnostic HMI

Project 1 - design and implementation

Representative I/O and tags

Tag	Type	Purpose
Start_PB	DI	Start request
Stop_OK	DI	Fail-safe stop chain
PE_Entry	DI	Object detection
M1_RunFb	DI	Running feedback
M1_RunCmd	DO	Contactor/VFD run
M1_SpdRef	AO	Speed reference
Batch_Count	INT	Accepted object count

Phased build plan

Phase	Dates	Exit deliverable
1 Design	5-11 Oct	Process sketch, I/O, tags, acceptance criteria
2 Build	12-25 Oct	Core logic, modes, sequence and comments
3 Integrate	26 Oct-8 Nov	HMI, VFD model, alarms and diagnostics
4 Prove	9-15 Nov	FAT, five faults, defect closure and demo

Required fault injections

- ☐ Blown control fuse
- ☐ Sensor LED on but PLC input off
- ☐ Repeated counts from maintained input
- ☐ Downstream Ready false
- ☐ Command true but run feedback absent
- ☐ VFD in LOCAL
- ☐ Reset attempted before correction

TESTING RULE

Write the expected result before each test. Record actual result, status, evidence ID, defect number and retest outcome. Keep unperformed tests marked NOT TESTED.

Project 1 - evidence, demonstration and grading

PORTFOLIO EVIDENCE

- ☐ README and one-page architecture
- ☐ Electrical/control schematic
- ☐ I/O/tag list
- ☐ Control narrative and sequence chart
- ☐ Commented PLC source/screenshots
- ☐ HMI views
- ☐ Cause/effect and alarm list
- ☐ FAT, defect log and fault reports
- ☐ Two-minute demonstration

STRETCH OPTIONS

- ☐ First-out fault display
- ☐ Reusable motor function block
- ☐ Trend command versus feedback
- ☐ Rockwell terminology comparison note

Demonstration sequence

1. Explain process and architecture
2. Trace one input and one output
3. Run a normal Auto cycle
4. Change mode and prove ownership
5. Inject a jam or feedback fault
6. Diagnose, correct and verify recovery

Project-specific acceptance gate

- ☐ All tags match drawings, logic and HMI
- ☐ Normal operation passes
- ☐ Stop and loss-of-permissive behavior passes
- ☐ At least five relevant faults are diagnosed
- ☐ Recovery requires correction before reset
- ☐ Backups and revision log exist
- ☐ Demo can be repeated without hidden setup

INTERVIEW LANGUAGE

Describe exactly what was simulated, configured, implemented, tested and documented. Separate introductory exposure from hands-on competence and never present the project as commissioned plant experience.

Project 2 - Automated Tank, Pump & Instrumentation Skid

BUILD WINDOW	PROJECT LABEL
16 Nov - 27 Dec 2026	Simulation / Training Project

PROJECT BRIEF

Simulate a fill, transfer and drain skid with 4-20 mA instruments, duty/standby pumps, control valves, alarms, trips, Manual/Automatic behavior and basic PID observation.

Learning objectives

- ☐ Select and scale realistic instruments
- ☐ Implement duty/standby and abnormal-signal handling
- ☐ Integrate PLC, process, HMI alarms and trends
- ☐ Use cause/effect and FAT to prove protection

Platforms and tools

PREFERRED	ACCESS RULE
☐ TIA Portal + S7-PLCSIM	☐ Confirm software availability before committing
☐ Ignition	☐ Use official documentation
☐ Factory I/O or software process model	☐ Record version used
☐ diagrams.net	☐ Use a documented fallback if unavailable
☐ Spreadsheet/document tools	☐ Do not overstate vendor competence

Functional requirements

- ☐ Tank level and outlet flow analog signals
- ☐ Two transfer pumps with duty alternation
- ☐ Inlet and outlet valves with position feedback
- ☐ Low-low permissive and High-High trip
- ☐ Manual/Automatic level behavior
- ☐ PID fundamentals observation only
- ☐ Instrument range, quality and plausibility alarms

Project 2 - design and implementation

Representative I/O and tags

Tag	Type	Purpose
LT_101	AI	0-100% level, 4-20 mA
FT_101	AI	0-1000 L/min
P101_RunFb	DI	Duty pump feedback
XV101_OpenFb	DI	Valve open position
P101_RunCmd	DO	Pump command
XV101_OpenCmd	DO	Inlet valve command
LIC101_Out	REAL	Controller output

Phased build plan

Phase	Dates	Exit deliverable
1 Instrument design	16-29 Nov	P&ID, selection matrix, ranges and loops
2 Analog build	30 Nov-13 Dec	Scaling, validation and calibration records
3 Process control	14-20 Dec	Sequence, valves, alarms and trends
4 Prove	21-27 Dec	High-High/valve tests, FAT and demo

Required fault injections

- ☐ Open 4-20 mA loop
- ☐ Wrong PLC scale
- ☐ Frozen level
- ☐ Noisy signal
- ☐ Pump overload
- ☐ Valve command/position discrepancy
- ☐ High-High during fill

TESTING RULE

Write the expected result before each test. Record actual result, status, evidence ID, defect number and retest outcome. Keep unperformed tests marked NOT TESTED.

Project 2 - evidence, demonstration and grading

PORTFOLIO EVIDENCE

- ☐ P&ID and process description
- ☐ I/O and instrument indexes
- ☐ Loop diagrams and scaling workbook
- ☐ Calibration record
- ☐ PLC/HMI evidence
- ☐ Alarm/cause-effect tables
- ☐ Trend interpretation
- ☐ FAT and three fault reports
- ☐ Three-minute demonstration

STRETCH OPTIONS

- ☐ Signal simulation panel
- ☐ Alarm rationalization note
- ☐ Controller trend comparison
- ☐ Maintenance faceplate

Demonstration sequence

1. Trace LT-101 end to end
2. Show 0/50/100 percent scaling
3. Run duty alternation
4. Inject valve discrepancy
5. Explain High-High protection
6. Recover and show records

Project-specific acceptance gate

- ☐ All tags match drawings, logic and HMI
- ☐ Normal operation passes
- ☐ Stop and loss-of-permissive behavior passes
- ☐ At least five relevant faults are diagnosed
- ☐ Recovery requires correction before reset
- ☐ Backups and revision log exist
- ☐ Demo can be repeated without hidden setup

INTERVIEW LANGUAGE

Describe exactly what was simulated, configured, implemented, tested and documented. Separate introductory exposure from hands-on competence and never present the project as commissioned plant experience.

Project 3 - SCADA/HMI, Historian & Industrial Communications

BUILD WINDOW	PROJECT LABEL
28 Dec 2026 - 24 Jan 2027	Simulation / Training Project

PROJECT BRIEF

Build a professional Ignition-based operator and diagnostic layer connected to simulated PLC data through OPC UA and a mapped Modbus device.

Learning objectives

- ☐ Design clear process and diagnostic displays
- ☐ Configure tags, quality, alarms, trends and history
- ☐ Create a valid IP/protocol plan
- ☐ Demonstrate layered communication fault finding

Platforms and tools

PREFERRED	ACCESS RULE
☐ Ignition	☐ Confirm software availability before committing
☐ OPC UA	☐ Use official documentation
☐ Modbus TCP simulator/device	☐ Record version used
☐ Windows networking tools + Wireshark	☐ Use a documented fallback if unavailable
☐ TIA Portal/PLCSIM data source	☐ Do not overstate vendor competence

Functional requirements

- ☐ Overview, area detail, pump faceplate and diagnostics
- ☐ At least 20 tags with type, scale and access
- ☐ Five prioritized alarms with operator response
- ☐ Live and historical trends
- ☐ OPC UA connection and Modbus map
- ☐ Bad-quality/stale-data behavior
- ☐ Read/write permissions and command confirmation
- ☐ Loss and recovery timing

Project 3 - design and implementation

Representative I/O and tags

Tag	Type	Purpose
Tank_Level	REAL R	Scaled PV
P101_Run	BOOL R	Pump feedback
P101_Start	BOOL RW	Confirmed command
P101_Fault	BOOL R	Alarm source
PLC_Watchdog	DINT R	Comms health
Data_Quality	QUALITY	Validity
Last_Update	TIME	Freshness

Phased build plan

Phase	Dates	Exit deliverable
1 UX design	28 Dec-3 Jan	Hierarchy, style guide and wireframes
2 Build	4-10 Jan	Ignition tags, views, alarms and history
3 Network	11-17 Jan	IP schedule, OPC UA and diagnostics
4 Protocol/FAT	18-24 Jan	Modbus, loss/recovery and demo

Required fault injections

- ☐ Wrong subnet
- ☐ Wrong Modbus register
- ☐ Wrong type/byte order
- ☐ Read-only command tag
- ☐ PLC offline
- ☐ Stale value shown as normal
- ☐ Excessive timeout

TESTING RULE

Write the expected result before each test. Record actual result, status, evidence ID, defect number and retest outcome. Keep unperformed tests marked NOT TESTED.

Project 3 - evidence, demonstration and grading

PORTFOLIO EVIDENCE

- ☐ Screen hierarchy and style rationale
- ☐ Architecture and IP schedule
- ☐ Tag/protocol register
- ☐ Alarm list and tests
- ☐ Trend/history evidence
- ☐ Command-access tests
- ☐ Loss/recovery report
- ☐ Seven fault reports
- ☐ Three-minute demonstration

STRETCH OPTIONS

- ☐ Communication-health dashboard
- ☐ Alarm performance summary
- ☐ Historian query/export
- ☐ Least-privilege role design

Demonstration sequence

1. Open overview and show state
2. Trace a tag from source to display
3. Acknowledge an alarm
4. Explain a historical event
5. Break mapping or communications
6. Diagnose by layer and recover

Project-specific acceptance gate

- ☐ All tags match drawings, logic and HMI
- ☐ Normal operation passes
- ☐ Stop and loss-of-permissive behavior passes
- ☐ At least five relevant faults are diagnosed
- ☐ Recovery requires correction before reset
- ☐ Backups and revision log exist
- ☐ Demo can be repeated without hidden setup

INTERVIEW LANGUAGE

Describe exactly what was simulated, configured, implemented, tested and documented. Separate introductory exposure from hands-on competence and never present the project as commissioned plant experience.

Project 4 - Offshore / Remote Process Package Troubleshooting & Commissioning Simulation

BUILD WINDOW	PROJECT LABEL
25 Jan - 28 Feb 2027	Simulation / Training Project

PROJECT BRIEF

Integrate an unattended or remotely operated pump/process package with robust local protection, remote SCADA, instrumentation, communications, commissioning evidence and a focused Rockwell vendor-translation exercise.

Learning objectives

- ☐ Write requirements and a safe-state philosophy
- ☐ Build reusable equipment modules and process sequences
- ☐ Retain local protection when SCADA fails
- ☐ Execute commissioning, defect and recovery tests
- ☐ Translate one Siemens pump subsystem into Rockwell concepts

Platforms and tools

PREFERRED	ACCESS RULE
☐ TIA Portal + PLCSIM primary	☐ Confirm software availability before committing
☐ Ignition SCADA/HMI/historian	☐ Use official documentation
☐ OPC UA + Modbus TCP	☐ Record version used
☐ Factory I/O or suitable process simulation	☐ Use a documented fallback if unavailable
☐ QElectroTech + diagrams.net	☐ Do not overstate vendor competence
☐ Studio 5000/Logix concepts for translation	

Functional requirements

- ☐ Remote pump, valve, level, pressure and flow package
- ☐ Local/Remote and Manual/Automatic ownership
- ☐ Start permissives, trips and reset philosophy
- ☐ Analog validation and discrepancy alarms
- ☐ SCADA quality, trends, history and command access
- ☐ Network-loss behavior and restoration
- ☐ FAT/SAT-style records, backups and handover

Project 4 - design and implementation

Representative I/O and tags

Tag	Type	Purpose
P101_Mode	ENUM	Ownership/mode
P101_PermOK	BOOL	Start permission
P101_Trip	BOOL	Latched protective stop
PT101	AI	Pressure 4-20 mA
FT101	AI	Flow 4-20 mA
Comms_OK	BOOL	Remote link health
Remote_Start	BOOL	Validated command

Phased build plan

Phase	Dates	Exit deliverable
1 Define	25 Jan-7 Feb	URS-lite, P&ID, documents and tests
2 Integrate	8-14 Feb	PLC, instruments, SCADA and network
3 Commission	15-21 Feb	I/O/loop checks, FAT, defects and retest
4 Release	22-28 Feb	Rockwell translation, final demo and handover

Required fault injections

- ☐ Pump fail-to-start
- ☐ Pressure loop open
- ☐ Low-flow with run feedback
- ☐ Valve discrepancy
- ☐ PLC-SCADA loss
- ☐ Stale historical value
- ☐ Wrong ownership
- ☐ Alarm flood after recovery
- ☐ Restore wrong PLC version

TESTING RULE

Write the expected result before each test. Record actual result, status, evidence ID, defect number and retest outcome. Keep unperformed tests marked NOT TESTED.

Project 4 - evidence, demonstration and grading

PORTFOLIO EVIDENCE

- ☐ Requirements and acceptance criteria
- ☐ P&ID, architecture and all registers
- ☐ PLC/HMI/SCADA evidence
- ☐ Network/protocol plan
- ☐ FAT/SAT and defect log
- ☐ Nine fault reports
- ☐ Backup/change/handover package
- ☐ Siemens-Rockwell comparison
- ☐ Five-to-ten-minute demonstration

STRETCH OPTIONS

- ☐ Requirements-to-test matrix
- ☐ First-out/event timeline
- ☐ Remote diagnostics page
- ☐ CompactLogix conceptual translation

Demonstration sequence

1. Explain safe-state and ownership
2. Trace a critical loop
3. Run normal remote operation
4. Lose SCADA and prove local protection
5. Close a defect and retest
6. Show Rockwell translation and limitations

Project-specific acceptance gate

- ☐ All tags match drawings, logic and HMI
- ☐ Normal operation passes
- ☐ Stop and loss-of-permissive behavior passes
- ☐ At least five relevant faults are diagnosed
- ☐ Recovery requires correction before reset
- ☐ Backups and revision log exist
- ☐ Demo can be repeated without hidden setup

INTERVIEW LANGUAGE

Describe exactly what was simulated, configured, implemented, tested and documented. Separate introductory exposure from hands-on competence and never present the project as commissioned plant experience.

Project 4 Rockwell vendor-translation exercise

Translate one Siemens-controlled pump subsystem, such as P-101, into Rockwell/Allen-Bradley concepts. The goal is conceptual transfer, not a second full capstone.

Required behavior

PUMP BEHAVIOR	ROCKWELL CONCEPTS
☐ Start/stop command	☐ CompactLogix/ControlLogix architecture
☐ Run feedback	☐ Studio 5000 project organization
☐ Overload input	☐ Controller and Program Tags
☐ Low-level permissive	☐ Tasks, Programs and Routines
☐ Trip and fault latch	☐ Ladder Diagram
☐ Manual/Automatic concept	☐ Timer and latch/unlatch concepts
☐ Status tags for HMI	☐ EtherNet/IP awareness

Required comparison

Siemens	Rockwell concept
S7-1200 / S7-1500	CompactLogix / ControlLogix
TIA Portal	Studio 5000 Logix Designer
PLC tags / DB data	Controller Tags / Program Tags
OB / FC / FB	Tasks / Programs / Routines; AOI awareness
LAD	Ladder Diagram
Profinet	EtherNet/IP awareness
Siemens diagnostics	Rockwell online monitoring concepts

EVIDENCE

One-page architecture comparison; Rockwell tag list; small Ladder routine or clearly labelled design exercise; project-organization diagram; Profinet/EtherNet-IP note; reflection on what transferred and what still requires training.

Portfolio packaging and final release

FOLDER PATTERN

00_README | 01_Requirements | 02_Design | 03_PLC | 04_HMI_SCADA | 05_Network | 06_Testing | 07_Faults | 08_Demo
| 09_Backups

Evidence index fields

- ☐ Evidence ID and title
- ☐ Project, requirement and test link
- ☐ Revision and date
- ☐ Studied / Simulated / Implemented / Tested / Documented
- ☐ What the artifact proves
- ☐ Tool/version used
- ☐ Known limitation
- ☐ File path

Final competency position

By 28 February 2027, the strongest practical story should be Siemens TIA Portal/S7 plus Ignition, instrumentation, communications, testing and troubleshooting. The Rockwell exercise demonstrates vendor adaptability while remaining clearly introductory.

README OPENING STATEMENT

This is a personal Simulation / Training Project created to demonstrate junior automation and control-system skills. It was not commissioned on a live plant. Safety functions, field wiring and site behavior are represented only to the extent stated in the scope and limitations.